

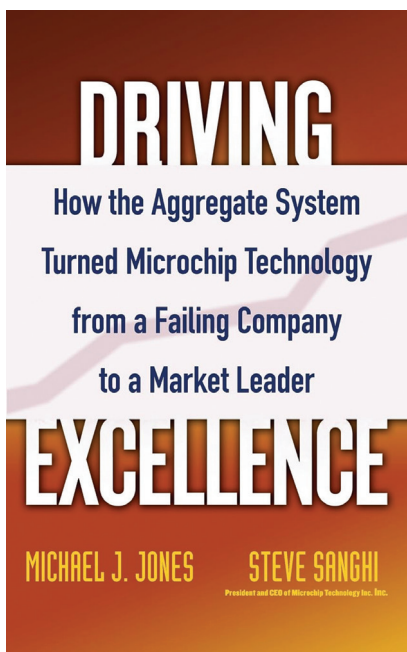
many of its products and expects it will become a significant component of our portfolio over time.

SG: India recently launched the India Semiconductor Mission with extensive incentives for semiconductor manufacturing. How would you assess India's progress so far?

SS: India's policy, including a 50% incentive for project costs, marks a promising start, but semiconductor manufacturing requires long-term commitment and consistency, often spanning decades. Policies must be stable and resilient to changes in government or economic shifts to attract foreign investment. The latest initiatives are more thoughtful than previous efforts, and the government's openness to adjusting policies based on industry feedback shows promise. However, establishing a semiconductor ecosystem that includes raw material supply chains and local support industries is crucial for India's success in semiconductor manufacturing.

SG: How do you see India's potential in compound semiconductors like gallium nitride (GaN) and silicon carbide (SiC), especially considering the demand for power applications in electric vehicles (EVs), renewables, and others?

SS: Compound semiconductors like GaN and SiC hold great promise, especially for high-power applications. Although GaN and SiC fabs require a smaller initial investment than silicon, the technology itself is challenging to master. However, with India's growing EV market, this could be a viable opportunity if Indian companies can overcome technical hurdles and gain customer trust. Global companies have faced challenges with compound semiconductors, so India should start small, focusing on local demand, and aim to build expertise gradually.



SG: With government support for the fabless ecosystem, what types of semiconductor products should Indian companies focus on for commercial viability?

SS: India's fabless ecosystem has two paths: one, innovate to address unmet global needs, or two, design products that cater to India's domestic market and gradually expand. An example of innovation is Microchip's early success with programmable microcontrollers, which enabled customer flexibility. Indian companies might initially focus on products with local demand, such as power management ICs for solar inverters, and later scale these products to international markets as they establish credibility.

SG: Turning to academic research, what advice would you offer to Indian institutions aiming to support India's semiconductor ambitions?

SS: Universities should bridge the gap between academic research and industry application. Too often, universities focus on theoretical research aimed at publications and

patents, whereas industry needs applied research that drives commercialisation. Sponsoring specific projects aligned with industry needs could enhance research relevance and lead to more direct commercial applications. Collaboration with industry, especially in design-focused research, will help produce commercially viable results.

SG: What is your recipe for building a company in India that could one day rival global giants like Microchip or MediaTek?

SS: Building a semiconductor giant requires long-term consistency, resilience, and a commitment to quality and reliability over decades. It starts with leadership at the top, paired with dedicated engineers and skilled labour. Consistency in policy and industry support is crucial, and fostering talent through training and collaboration with industry will help lay the foundation. My hope is that India can create a globally competitive semiconductor company within the next few decades—one with the tenacity to endure cycles of growth and recession alike.

SG: Any closing thoughts for those working towards India's semiconductor ambitions?

SS: India has made significant strides in a brief time, and people like yourself, the VLSI Society, and institutions are all contributing to a promising ecosystem. My advice is to celebrate these accomplishments but remain aware of the room for improvement, much like how I encouraged my children to aim high but stay grounded. With continued dedication, India's semiconductor sector will have a strong foundation for success. My best wishes for VLSI Society for the upcoming 38th VLSI Design conference in January 2025 which Microchip is supporting as an anchor company. **EFY**